Design Your Experiment Experiment 3

You are going to design an experiment about differential amplifiers.

1. Your experiment must be different than the Experiment-2.
2. Circuit is going to amplify a sinusoidal input signal generated by a function generator.
3. The input and output signals are going to be observed on the oscilloscope.
4. A report will be submitted, which will include your set-up, measurements and the SPICE simulations.
5. It is strongly suggested to design your experiment first in SPICE.
6. You can use the components listed below. You may also use other components and equipment in the lab.

Equipment/Component	**Model/Value**	**Max Count**
Breadboard	-	1
Resistor	2.2kΩ	2
Resistor	100Ω	6
Resistor	220Ω	2
Resistor	330Ω	2
Resistor	750Ω	2
Resistor	820Ω	2
Resistor	1kΩ	2
Resistor	10kΩ	2
Resistor	15kΩ	2
BJT (npn)	BC547	4
BJT (pnp)	BC556	4
Multi-meter	-	1
Oscilloscope	-	1
Function Generator	Sine Wave	1
Voltage Source	Adjustable	1
Probes & Cables	-	-

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Pre-LAB Work

1. For the circuit given in Fig. 1, calculate the differential mode (Vout/Vid) and common mode (Vout/Vcm) voltage gains. Ignore the Early Effect and assume that all the transistors are operating in active mode. Vcc=15V and Vee=-15V.

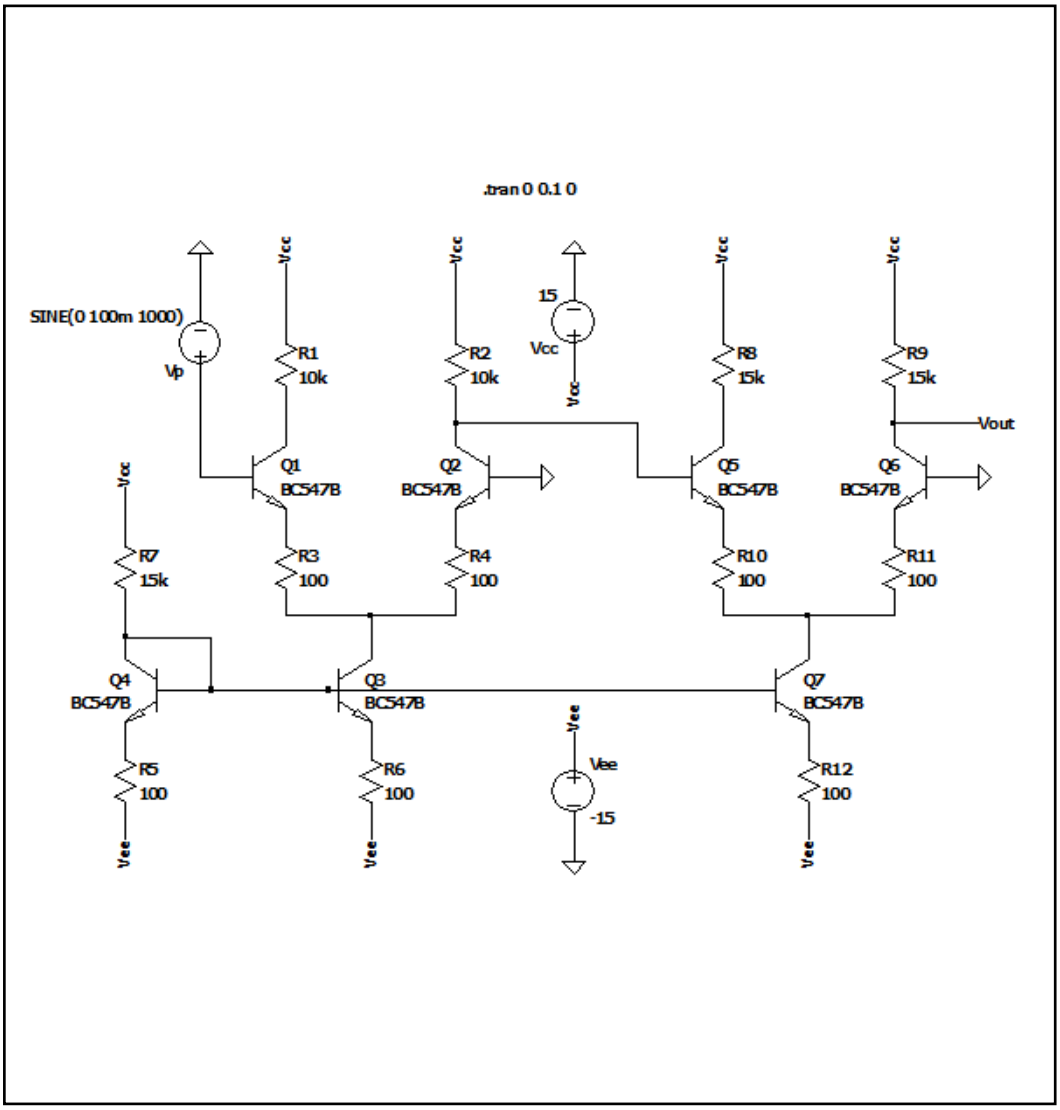
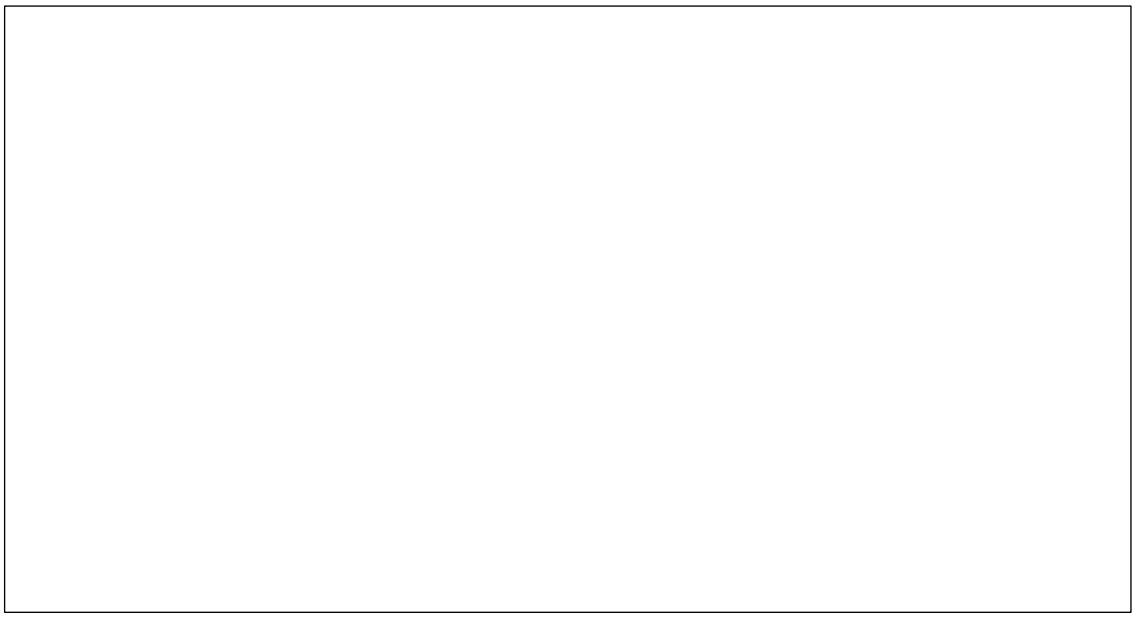


Figure 1



2. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Apply a sine wave of 1kHz with an amplitude of 100mV to Q2 and Q6 to ground. Measure the differential mode voltage gains.

Ad=(Vout[peak-to-peak]/100mV)= 13.88V/100mV=138

Equipment/Component List

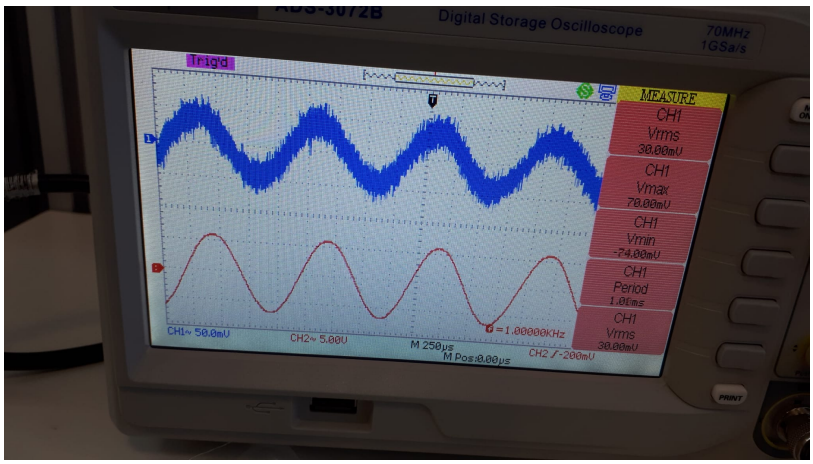
Table 1: Equipment/Component List

Equipment/Component	**Model/Value**	**Count**
Breadboard	-	1
Resistor	100Ω	**7**
Resistor	10kΩ	**2**
Resistor	15kΩ	3
BJT (npn)	BC547	**7**
Multi-meter	-	1
Voltage Source	15V, -15V	1
Probes & Cables	-	-

Experiment Procedure

4. Adjust Vcc to 15V and Vee to -15V.
5. Set-up the circuit in Fig. 1.
6. Apply a sine wave of 1kHz with an amplitude of 100mV to Q1.
7. Connect Q2 and Q6 to ground.
8. Plot Vout and measure the peak-to-peak value of the output.

Plot:



9. Calculate the differential mode voltage gain.

Ad=(Vout[peak-to-peak]/100mV)=10V/100mV=100

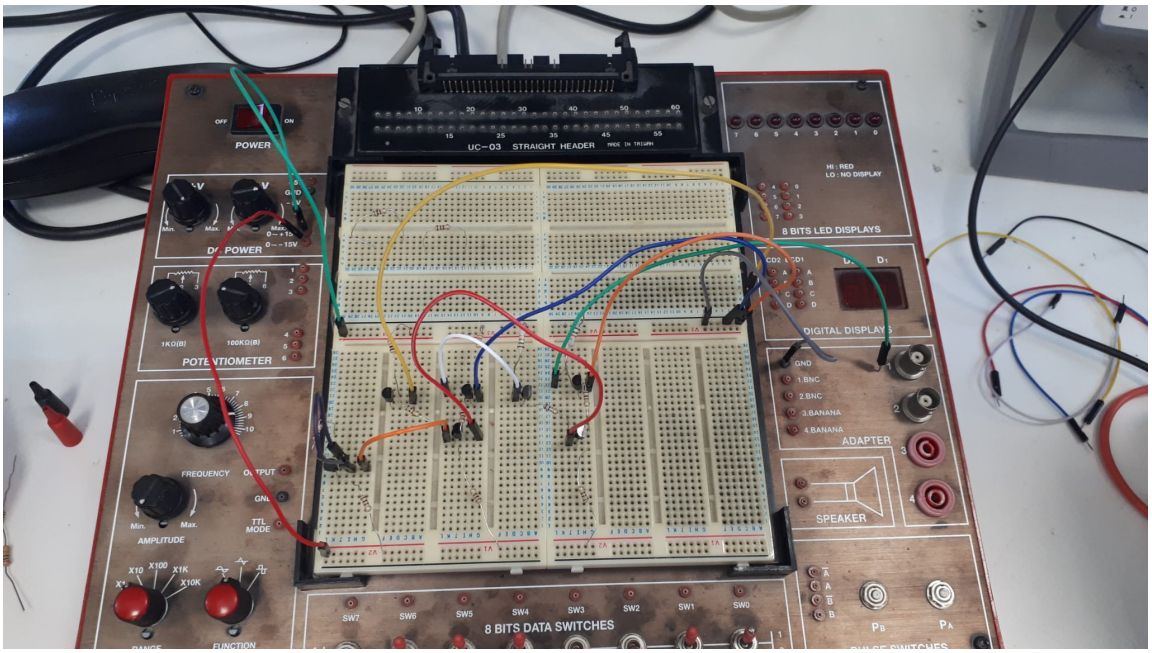


Figure 1: Constructed circuit in lab.

12. Compare the simulated (2) and measured (9) differential mode voltage gains.

Answer:

In this experiment the reason of designing the circuit, amplifying a sinusoidal input signal generated by a function generator. While we design the circuit we used npn transistors. After the circuit has been designed , we gave input voltage of 1kHz 100m sine wave. Then we observed the signal on the oscilloscope. We designed the the same circuit on the Spice as well. Using the graphs we obtained from both simulations we calculated differential mode voltage gains by using the peak to peak output voltage.

Finally the values we obtained in the laboratory session and spice were compared with each other. The purpose of this experiment is to observe amplified sinusoidal input signal generated by a function generator, converting the theoretical knowledge to real circuit. The peak-to-peak input voltage is 100mV and peak-to-peak output voltage is 10 V. We obtained the amplified sinusoidal output signal as a result. So when we compare the results from th spice with the measured result we can say that there is almost 38% of error. The fact that the resistors we use are not exactly the same may have been caused by voltage losses in the cables and errors in the measuring instruments.

As the main idea of the experiment, we can observe the output voltage, which is about 100 times larger than a small or even a very small input voltage that we have applied to the circuit.

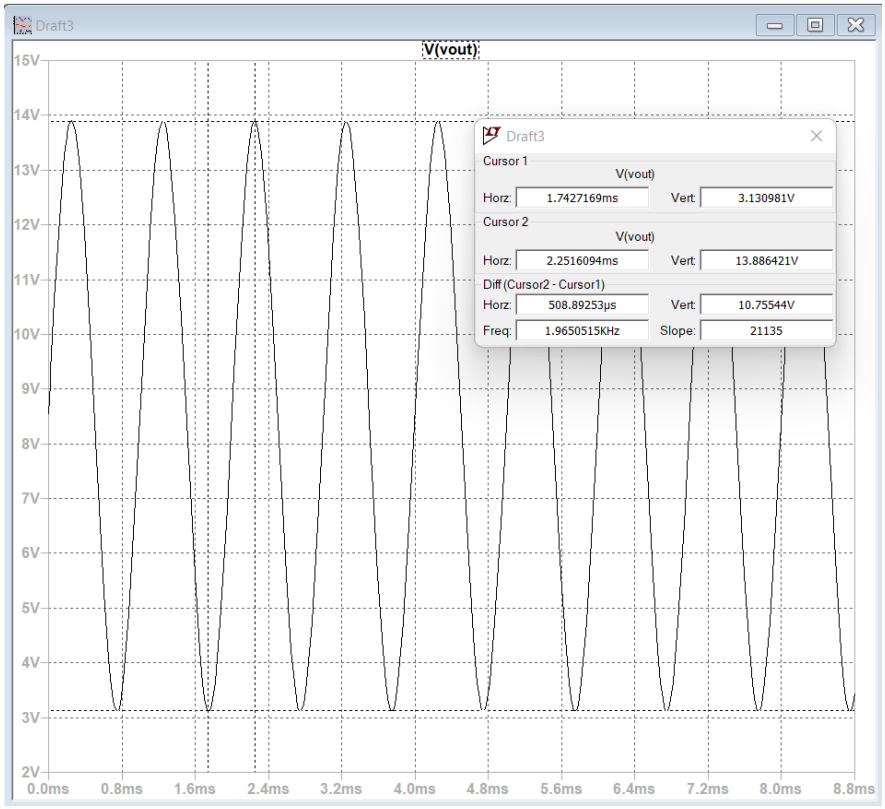


Figure 2: Ltspice Simulation result